

2007

**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet, formulae sheets and Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 9, 13, 17, 19, 21 and 25

Total marks – 100

Section I Pages 2–26

75 marks

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–27
- Allow about 1 hour and 45 minutes for this part

Section II Pages 27–40

25 marks

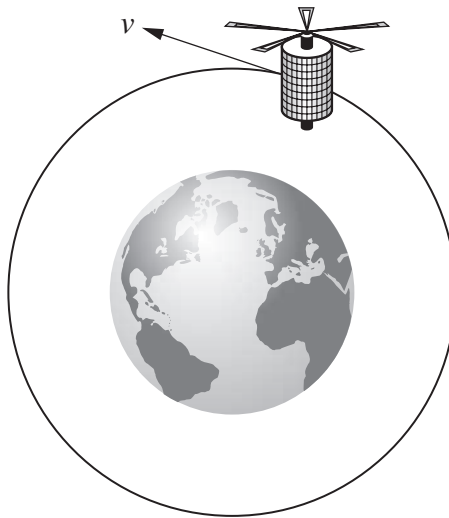
- Attempt ONE question from Questions 28–32
- Allow about 45 minutes for this section

Section I
75 marks

Part A – 15 marks
Attempt Questions 1–15
Allow about 30 minutes for this part

Use the multiple-choice answer sheet for Questions 1–15.

- 1** A satellite is in orbit around Earth with tangential velocity v as shown.



Which of the following describes the direction of the centripetal force acting on the satellite?

- (A) Same direction as the gravitational force
- (B) Opposite direction to the gravitational force
- (C) Same direction as the tangential velocity
- (D) Opposite direction to the tangential velocity

- 2 A spaceship sitting on its launch pad is measured to have a length L . This spaceship passes an outer planet at a speed of $0.95c$.

Which observations of the length of the spaceship are correct?

	<i>Observer on the spaceship</i>	<i>Observer on the planet</i>
(A)	No change	Shorter than L
(B)	No change	Greater than L
(C)	Shorter than L	No change
(D)	Greater than L	No change

- 3 The gravitational potential energy of a given mass is known at both Earth's surface and at a fixed distance above Earth.

What CANNOT be determined by comparing these two values of gravitational potential energy?

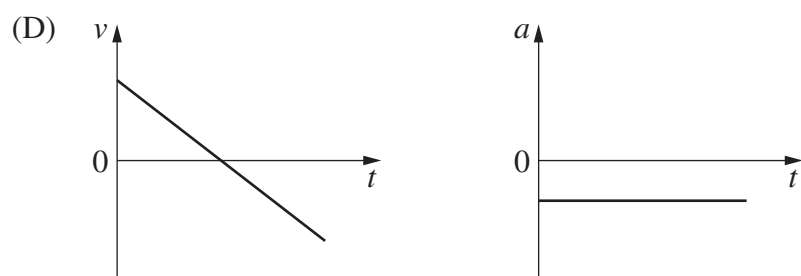
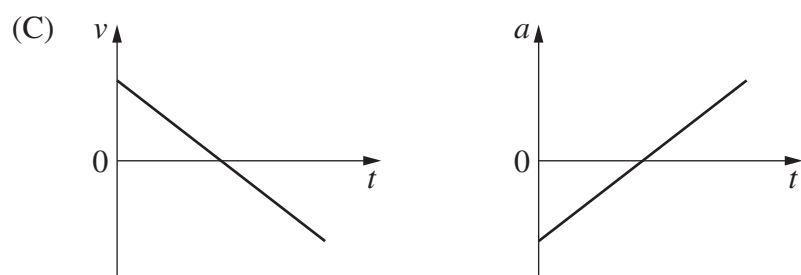
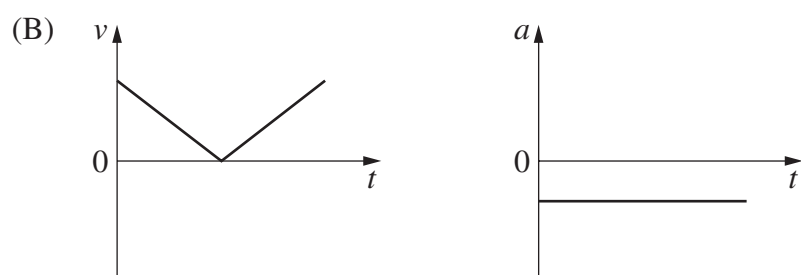
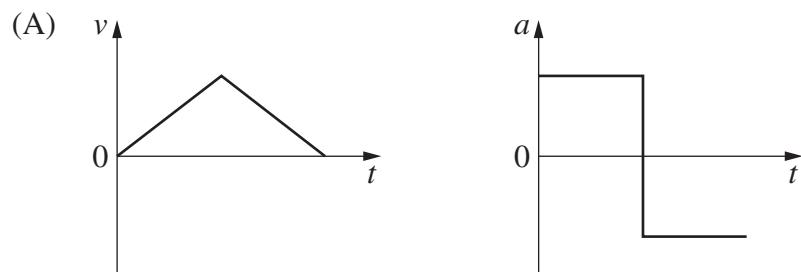
- (A) The mass of Earth
(B) The speed of rotation of Earth
(C) The escape velocity of a satellite from Earth
(D) The work done in moving between the two points
- 4 The acceleration due to gravity on Earth's surface is g . Suppose the radius of Earth was reduced to a quarter of its present value while its mass remained the same.

What would be the new value of the acceleration due to gravity on the surface?

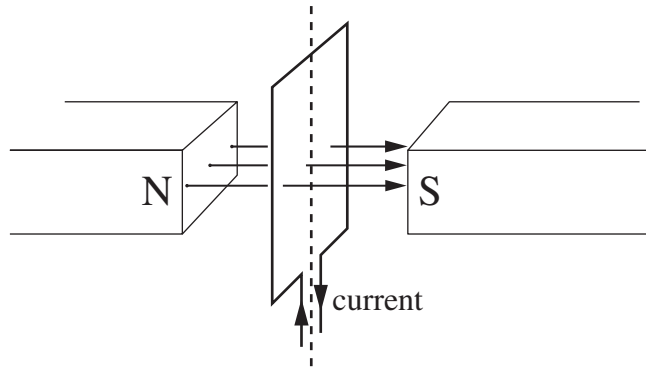
- (A) $\frac{1}{16}g$
(B) $\frac{1}{4}g$
(C) $4g$
(D) $16g$

- 5 A cannon ball is fired vertically upward from a stationary boat.

Which pair of graphs best describes the velocity, v , and acceleration, a , of the cannon ball as functions of time, t ? Ignore air resistance.



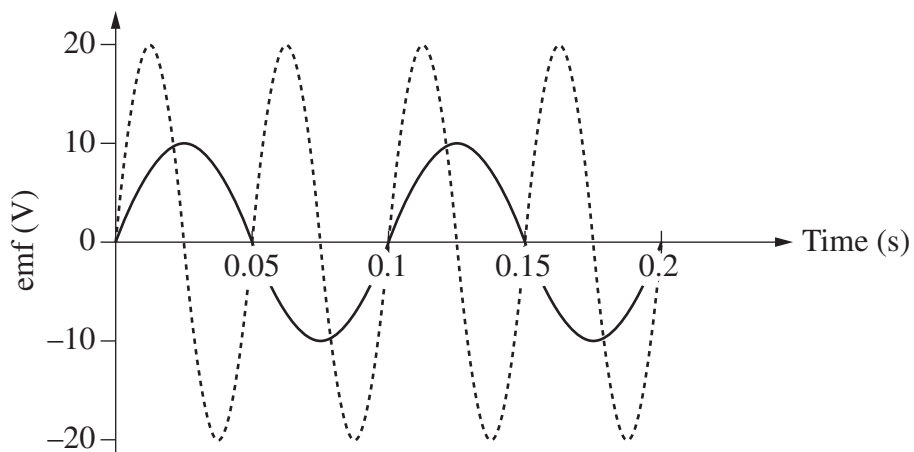
- 6 An electric motor is set up as shown.



When current is supplied the coil does not turn.

Which of the following is required for the coil to start turning?

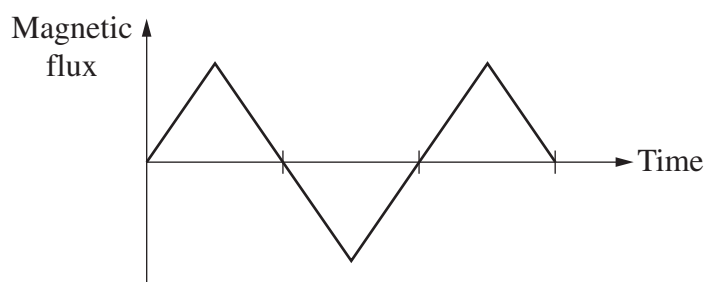
- (A) The magnetic field must be increased.
 - (B) The direction of the current must be reversed.
 - (C) The magnitude of the current must be increased.
 - (D) The starting position of the coil must be changed.
- 7 In the graph shown, the solid curve shows how the emf produced by a simple generator varies with time. The dashed curve is the output from the same generator after a modification has been made to the generator.



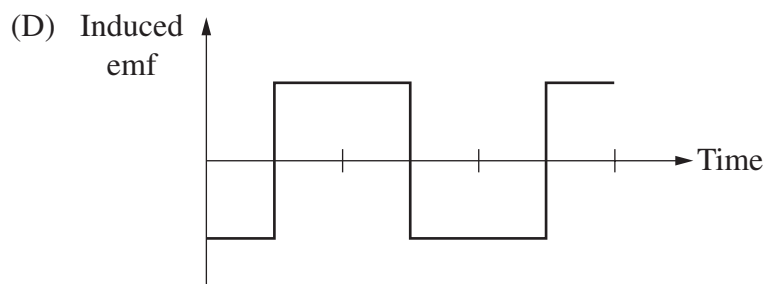
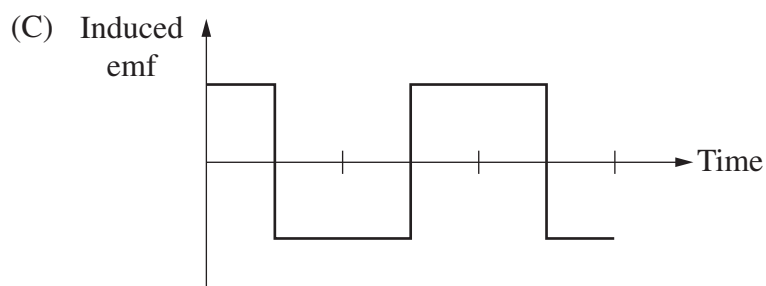
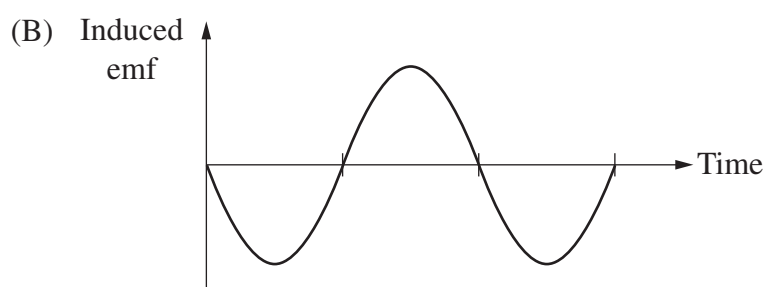
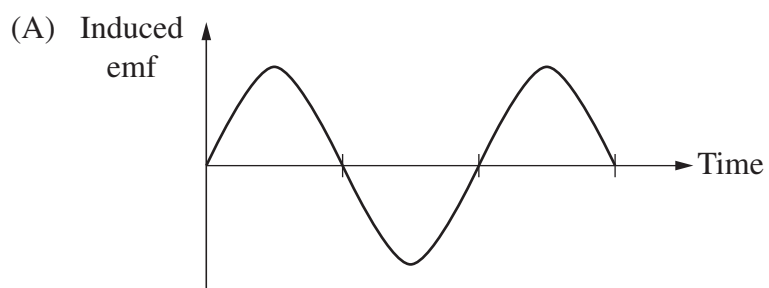
Which modification was made to produce the result shown?

- (A) The area of the coil was doubled.
- (B) A split-ring commutator was added.
- (C) The speed of rotation of the coil was doubled.
- (D) The number of turns in the coil was quadrupled.

- 8 The variation in magnetic flux through a coil is shown below.



Which graph best represents the corresponding induced emf in the coil?

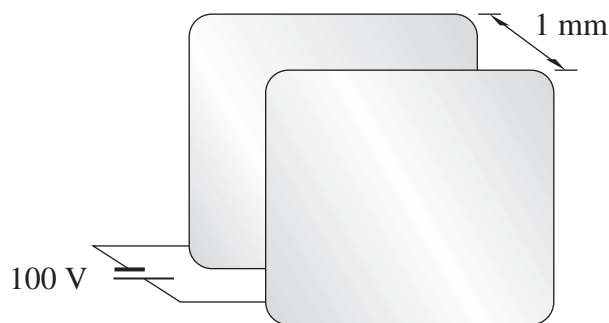


- 9 A stationary exercise bike has a solid metal wheel that is rotated by a chain connected to the pedals. An array of strong permanent magnets provides a magnetic field close to the face of the wheel.

The exercise level can be selected from 1 (easy) to 6 (hard) using a control panel.

When level 6 is selected, which of the following statements is correct?

- (A) The current supplied to the bike is a minimum.
 - (B) The magnetic field at the wheel is a minimum.
 - (C) The induced current in the wheel is a maximum.
 - (D) The distance between the magnets and the wheel is a maximum.
- 10 What is the wave property that enabled Hertz to calculate the velocity of radiowaves and compare it to the velocity of light?
- (A) Interference
 - (B) Polarisation
 - (C) Reflection
 - (D) Refraction
- 11 Two parallel metal plates are 1 mm apart. A potential difference of 100 V is applied as shown.



What is the magnitude of the uniform electric field between the plates?

- (A) 10^{-3} V m^{-1}
- (B) 10^{-1} V m^{-1}
- (C) 10^2 V m^{-1}
- (D) 10^5 V m^{-1}

- 12** The Bragg experiment used X-rays to investigate crystal structure.

Which statement best describes the results of this experiment?

- (A) X-rays are scattered from a crystal and form an interference pattern.
- (B) X-rays penetrate a crystal and form an interference pattern behind it.
- (C) X-rays are absorbed and re-emitted equally in all directions by a crystal.
- (D) X-rays absorbed by a crystal produce minima and those reflected produce maxima.

- 13** An electron is moving near a long straight wire. When a current is applied to the wire the electron experiences a force in the same direction as the current flow in the wire.

What was the electron's initial direction of motion?

- (A) Parallel to the current direction
- (B) Opposite to the current direction
- (C) Towards the wire and perpendicular to it
- (D) Away from the wire and perpendicular to it

- 14** Which of the following is a property of a silicon *p*-type semiconductor?

- (A) It is positively charged.
- (B) It has fewer holes than pure silicon.
- (C) It can be produced by doping the silicon with boron.
- (D) It can be produced by doping the silicon with phosphorus.

- 15** What CANNOT be calculated using the principle of conservation of energy?

- (A) The energy of a proton at rest
- (B) The production of back emf in a motor
- (C) Voltage transformation in an ideal transformer
- (D) The escape velocity of an object from the gravitational field of a planet

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Centre Number

Section I (continued)

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Part B – 60 marks

Student Number

Attempt Questions 16–27

Allow about 1 hour and 45 minutes for this part

Answer the questions in the spaces provided.

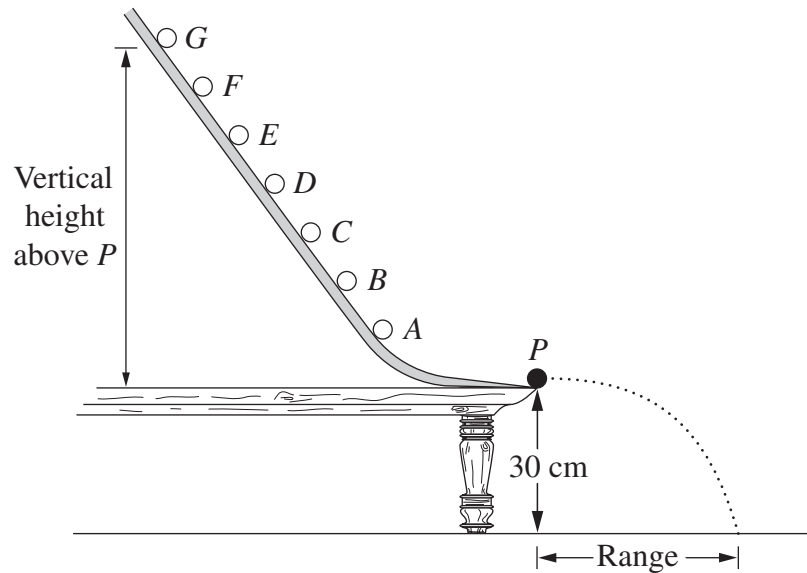
Show all relevant working in questions involving calculations.

Question 16 (5 marks)

Please turn over

Question 16 (5 marks)

A group of students conducted an investigation in which ball bearings were released from various points on a ramp. The ball bearings rolled down the ramp to the edge of a table at point *P* as shown. Their ranges were measured.



The results are shown in the table.

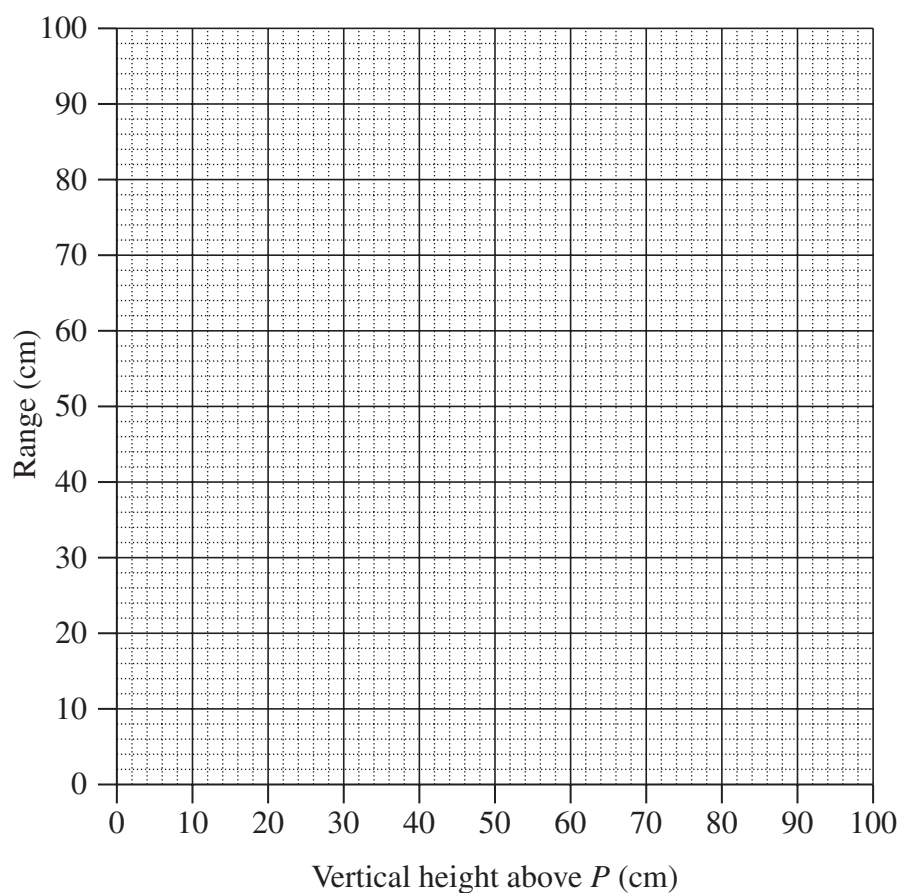
<i>Point of release</i>	<i>Vertical height above P (cm)</i>	<i>Range (cm)</i>
<i>A</i>	10	32
<i>B</i>	20	44
<i>C</i>	30	58
<i>D</i>	40	66
<i>E</i>	50	76
<i>F</i>	60	82
<i>G</i>	70	87

Question 16 continues on page 11

Question 16 (continued)

- (a) Plot the data from the table and draw a curve of best fit.

2



- (b) (i) Using your graph, predict the range of a ball bearing released from a height of 80 cm above point P .

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- (ii) Calculate the horizontal velocity of the ball bearing released from a height of 80 cm above point P .

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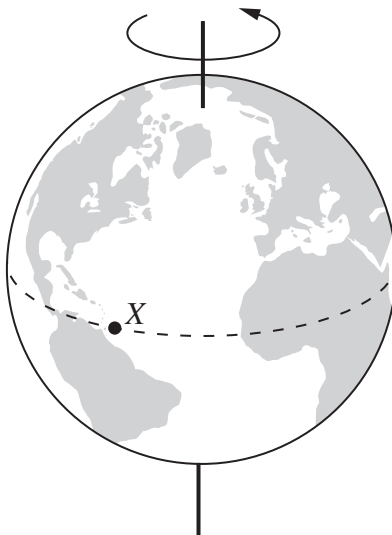
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End of Question 16

Question 17 (4 marks)

The diagram shows the position X on Earth's surface from which a satellite is to be launched into a geostationary orbit.



- (a) On the diagram, draw an arrow to show the direction of launch from X , and justify your choice. **1**

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- (b) Given that the radius of Earth is 6.38×10^6 m, calculate the height of the satellite above Earth's surface. **3**

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Centre Number

Section I – Part B (continued)

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Student Number

Marks

Question 18 (7 marks)

- (a) How has our understanding of time been influenced by the discovery of the constancy of the speed of light? 2

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- (b) A piece of radioactive material of mass 2.5 kilogram undergoes radioactive decay. How much energy is released if 10 grams of this mass are converted to energy during the decay process? 2

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- (c) A mass is moving in an inertial frame of reference at a velocity v relative to a stationary observer. The observer measures an apparent mass increase of 0.37%. 3

Calculate the value of v in m s^{-1} .

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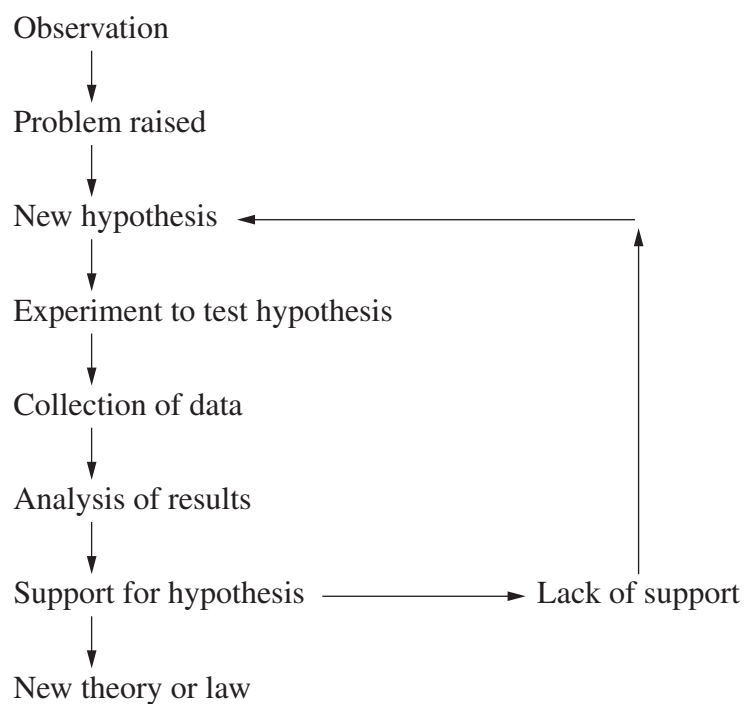
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Question 19 (6 marks)

This flowchart represents one model of scientific method used to show the relationship between theory and the evidence supporting it.



Question 19 continues on page 15

Question 19 (continued)

Analyse Einstein's Theory of Special Relativity and the evidence supporting it as an application of this model of scientific method.

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End of Question 19

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Centre Number

Section I – Part B (continued)

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Student Number

Marks

Question 20 (4 marks)

Assess the effects of the development of AC generators on today's society and the environment.

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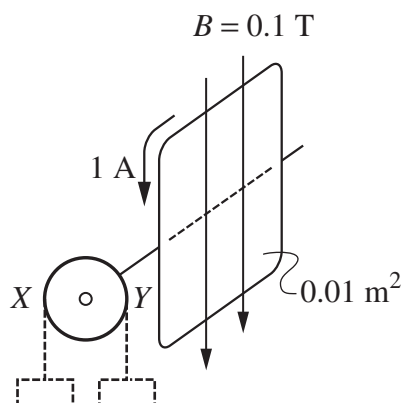
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Question 21 (5 marks)

A simple motor consists of a flat rectangular coil with n turns in a magnetic field B as shown.



The coil has an area of 0.01 m^2 and carries a current of 1 A . The motor drives a pulley of diameter 20 cm , and weights can be hung from either side of the pulley at point X or point Y .

- (a) In order to prevent rotation, should a weight be hung at point X or at point Y ? 1
Justify your answer.

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- (b) What is the magnitude of the torque provided by a mass of 0.2 kg suspended from either point X or point Y ? 2

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- (c) If the motor is just stopped by a mass of 0.2 kg , how many turns does the coil have? 2

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Centre Number

Section I – Part B (continued)

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Student Number

Marks

Question 22 (4 marks)

Explain why solid state devices have largely replaced thermionic devices.

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Please turn over

Question 23 (4 marks)

- (a) The table shows the critical temperature T_c at which some materials become superconducting. 1

<i>Year discovered</i>	<i>Material</i>	T_c (K)	T_c (°C)
1941	Niobium nitride	16	−257
1987	YBa ₂ Cu ₃ O ₇ (YBCO)	92	−185
1993	HgBa ₂ Ca ₂ Cu ₃ O ₇	133	−140

With reference to the table, identify what scientists working in the area of superconductivity are trying to achieve.

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- (b) Describe what happens to one property of superconductors near the critical temperature. 1

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- (c) Explain why magnetic levitation occurs in superconducting materials. 2

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Centre Number

Section I — Part B (continued)

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Student Number

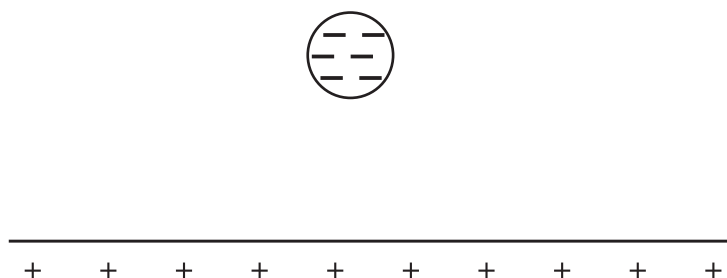
Question 24 (3 marks)

Please turn over

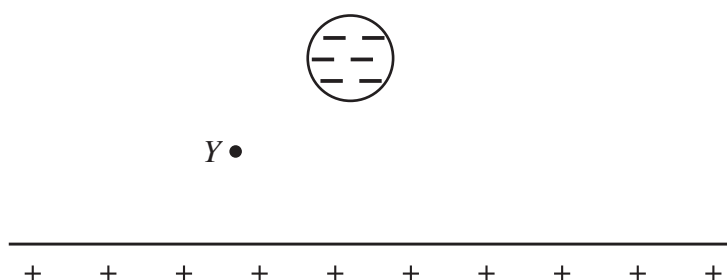
Question 24 (continued)

- (a) A negatively charged cylinder is fixed in position near a positively charged plate as shown in the cross-section. 1

Sketch the electric field lines between the cylinder and the plate on the cross-section diagram.



- (b) A tiny particle of mass 10^{-30} kg and charge $+6 \times 10^{-12}$ C is released at point Y as shown on the diagram. The particle initially accelerates at 7.0×10^{21} m s⁻². 2



Calculate the electric field intensity at Y.

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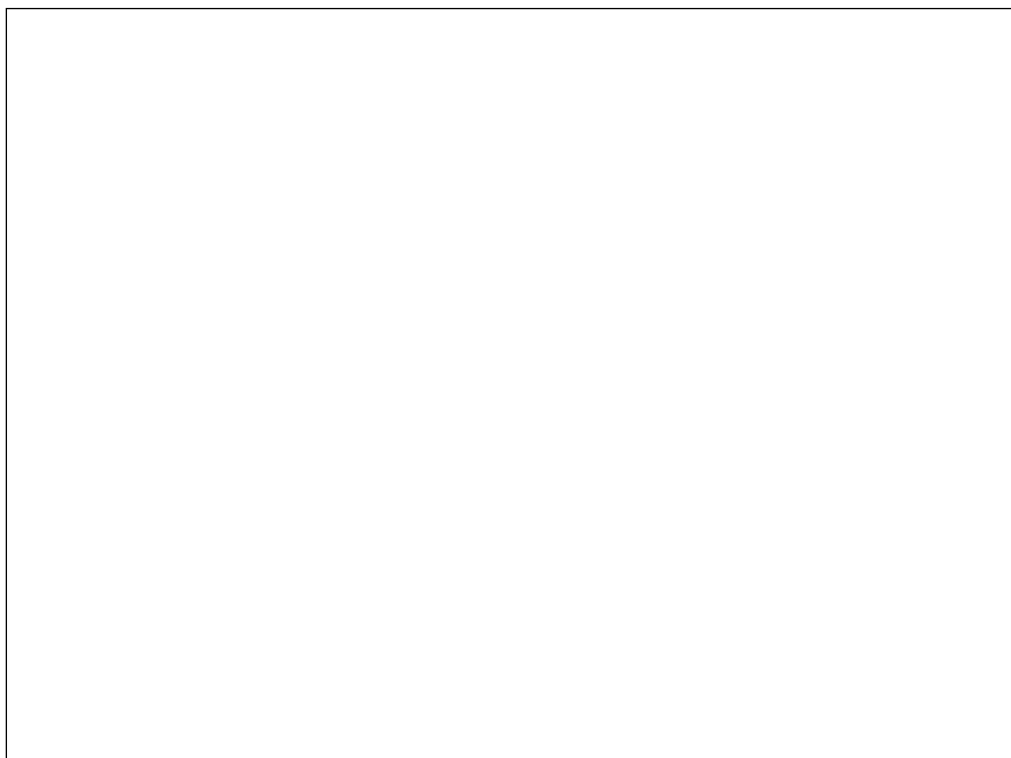
Question 25 (4 marks)

A student claims that a DC generator is an ‘electric motor in reverse’.

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Analyse this claim with reference to the structure and function of a simple DC generator and an electric motor.

Include diagrams in your answer.



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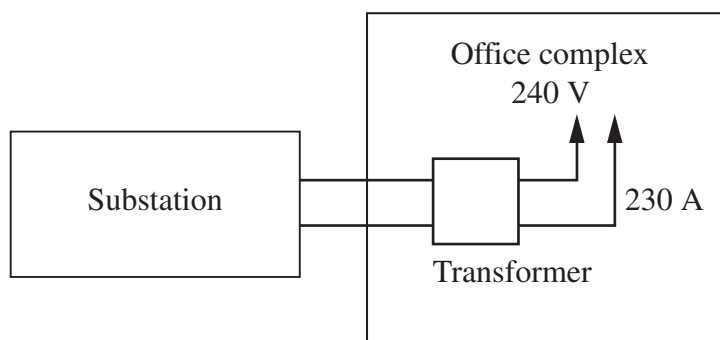
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Question 26 (6 marks)

An electricity substation delivers a current of 10 A at a voltage of 6 kV to an office complex. The office complex uses a transformer to provide a current of 230 A at a voltage of 240 V.



- (a) Explain why AC is preferable to DC as an input current for transformers. 2

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- (b) Outline possible causes of energy loss in the transformer. 2

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- (c) Calculate the energy lost by the transformer in eight hours. 2

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Centre Number

Section I – Part B (continued)

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Student Number

Marks**Question 27** (8 marks)

- (a) Scientists tried to explain observations of black-body radiation using classical wave theory and then quantum theory. **3**

How does quantum theory satisfactorily explain black-body radiation?

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- (b) Describe Hertz's observation of the photoelectric effect. **2**

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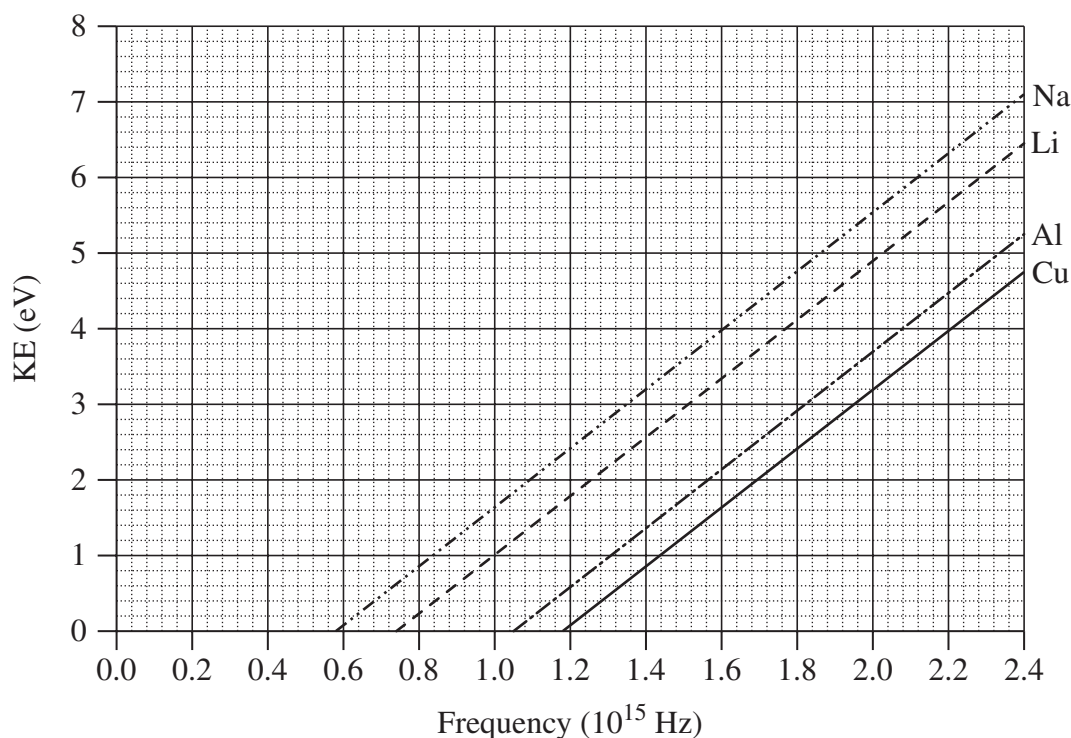
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Question 27 continues on page 26

Question 27 (continued)

- (c) An experiment was conducted in which light of different frequencies was shone onto the surfaces of four different metals. Electrons were found to be emitted and their kinetic energies were measured. The graph shows the results.



- (i) Calculate the gradient of the sodium (Na) graph. Show your working. 2

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- (ii) Each of the graphs has the same gradient. What is the significance of this observation? 1

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End of Question 27

Physics

Section II

25 marks

Attempt ONE question from Questions 28–32

Allow about 45 minutes for this section

Answer the question in a writing booklet. Extra writing booklets are available.

Show all relevant working in questions involving calculations.

	Pages
Question 28 Geophysics	28–30
Question 29 Medical Physics	31–33
Question 30 Astrophysics	34
Question 31 From Quanta to Quarks	35–37
Question 32 The Age of Silicon	38–40

Question 28 — Geophysics (25 marks)

- (a) (i) The graph shows the velocity of P and S waves at different depths in the Earth. **3**



Describe the rigidity and density of rocks at different depths in the Earth and relate this to the velocity of both P and S waves.

- (ii) The velocity of S waves through rock is given by the formula **2**

$$v_s = \sqrt{\frac{\mu}{\rho}} \quad \text{where } \mu = \text{rigidity} \\ \rho = \text{density.}$$

Two rocks, A and B , are compared. Both have the same rigidity but rock A has twice the density of rock B . If the velocity of S waves in rock B is 7 km s^{-1} , calculate the velocity of S waves in rock A .

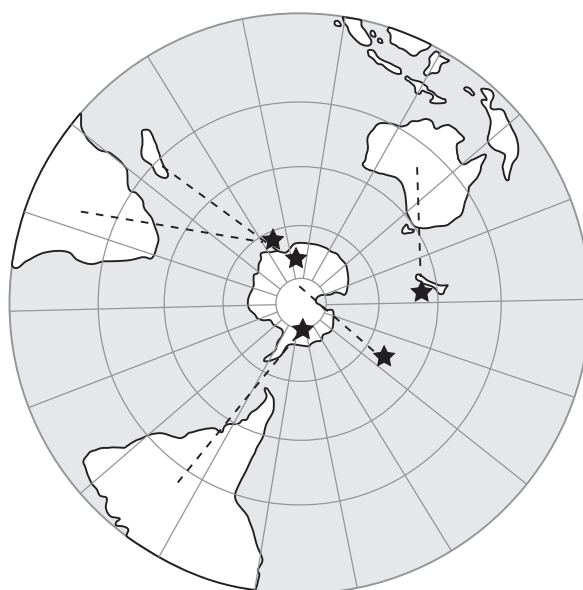
Question 28 continues on page 29

Question 28 (continued)

- (b) (i) In your study of Geophysics you performed an investigation that models the variation of the inclination of Earth's magnetic field with latitude. 2

Describe how you modelled the variation and the results obtained.

- (ii) The map below shows the positions of the south magnetic pole about 200 million years ago as measured from each of five present day land masses. The positions of the south magnetic pole are shown as ★ on the map. 4



Explain how these south magnetic pole positions were determined and how they provide evidence for continental drift.

- (c) Remote sensing by satellites has added greatly to our ability to understand Earth. 7

Analyse how the introduction of satellites has contributed to TWO areas of knowledge of Earth that were previously inaccessible.

Question 28 continues on page 30

Question 28 (continued)

- (d) Seismograms are shown for a nuclear test and an earthquake.



Ack: Lawrence Livermore National Laboratory

- | | | |
|-------|---|----------|
| (i) | By using features of the seismograms, compare the nature of the energy release during a nuclear test and an earthquake. | 3 |
| (ii) | What information do seismograms provide about nuclear tests? | 2 |
| (iii) | Describe how seismogram studies are used in either natural hazard reduction or mineral exploration. | 2 |

End of Question 28

Question 29 — Medical Physics (25 marks)

- (a) Ultrasound imaging depends on the fact that different materials have different acoustic impedances. The density and acoustic impedance of fat and muscle are shown.

	Density (kg m^{-3})	Acoustic impedance ($\text{kg m}^{-2} \text{s}^{-1} \times 10^6$)
Fat	9.25×10^2	1.38
Muscle	1.073×10^3	1.70

- (i) Calculate the fraction of ultrasound intensity that is transmitted as it passes from fat into muscle. 2
- (ii) Explain how a piezoelectric crystal can act as both a source and detector of ultrasonic waves. 3
- (b) (i) In your study of Medical Physics you performed an investigation to identify the function of the components of MRI equipment. The following information is from the *Australian Twin Registry Newsletter*, 2006.

MRI: How does it work?

Unlike conventional radiography Computed Tomography (CT) scans, which make use of potentially harmful X-rays passing through a patient to generate images, MRI is based on the magnetic properties of atoms within the tissue. A powerful magnet generates a magnetic field roughly 10,000 times stronger than the Earth's. A very small percentage of hydrogen atoms within the body will line up with this field. Radio waves are passed through the person's body then "bounce back" depending on how many lined up atoms they meet. The subtle differences in how the radio waves bounce back provide the information needed to construct the image.

Australian Twin Registry

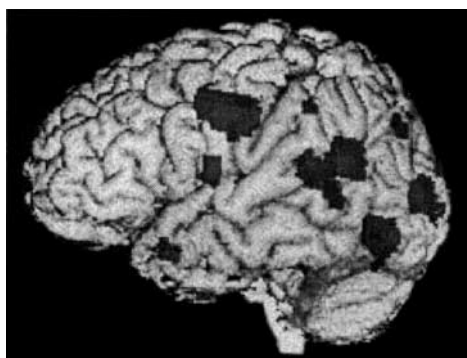
- Assess the information provided in this article in terms of its scientific accuracy. 4
- (ii) Why is ultrasound imaging NOT used for investigating the function and structure of the brain? 2

Question 29 continues on page 32

Question 29 (continued)

- (c) Assess the effects on the lives of individuals and the wider community of the production and use of X-rays, CAT scans and MRI. **7**
- (d) (i) Compare the advantages and disadvantages of CAT and PET scans in investigating the structure and function of the brain with reference to the images shown. **3**

Image A



Dark areas show abnormal brain activity function.

Image B



White area shows the result of a stroke.

Question 29 continues on page 33

Question 29 (continued)

Osteoporosis is a degenerative disease in which there is a gradual loss of minerals in the bone. This leads to decreased bone strength and an increased chance of bone fractures.



- (ii) With reference to the images shown, compare TWO imaging techniques which can be used to diagnose osteoporosis. **4**

End of Question 29

Question 30 — Astrophysics (25 marks)

- (a) (i) Describe how the existence of a binary star may be deduced. **2**
- (ii) The distance from the Sun to Earth is 1 astronomical unit (1 AU). The apparent magnitude of the Sun, when viewed from Earth, is -26.5 . Calculate the apparent magnitude of the Sun when viewed from Saturn, a distance of 10 AU from the Sun. **3**
- (b) (i) In your study of Astrophysics you performed a first-hand investigation to demonstrate ONE characteristic of a star that can be deduced by observing the light received from it. **2**
- Describe your investigation and the results obtained.
- (ii) Explain how TWO other characteristics or properties of a star can be deduced by observing the light received from the star. **4**
- (c) Evaluate the ways that TWO technologies used in modern astrophysics have changed scientific understanding of celestial objects. **7**

- (d) The table shows some characteristics of selected stars.

<i>Star</i>	<i>Apparent magnitude</i>	<i>Absolute magnitude</i>	<i>Spectral class</i>
Sun	-26.5	$+4.7$	G2
Sirius A	-1.51	$+1.4$	A1
Betelgeuse	$+0.41$	-5.6	M2
Arcturus	$+0.00$	-0.3	K2
Vega	$+0.04$	$+0.5$	A0

- (i) Which star is the most distant from Earth? Justify your answer. **1**
- (ii) Sketch in your writing booklet a Hertzsprung-Russell diagram showing the positions of the stars in the table. Identify the position of the Main Sequence on your diagram. **3**
- (iii) Compare the physical properties, nuclear reactions and relative stages of stellar evolution of the stars Sirius A and Arcturus. **3**

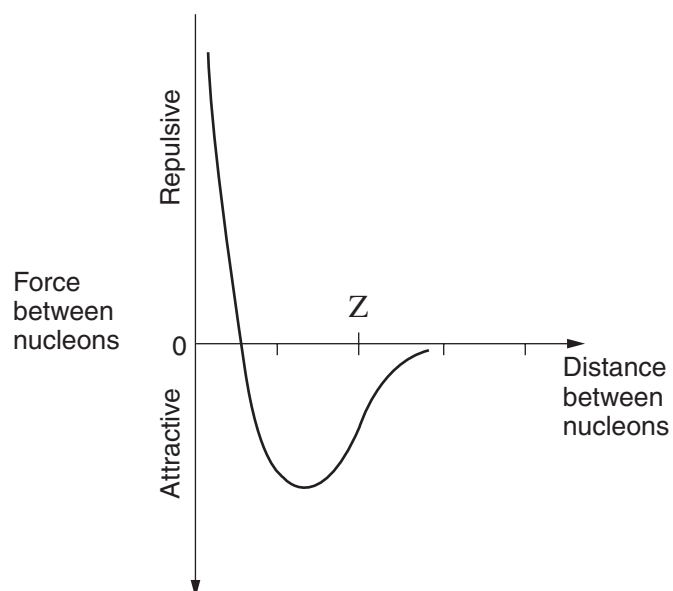
Question 31 — From Quanta to Quarks (25 marks)

- (a) The Bohr model of the atom describes a nucleus with electron shells surrounding it.
- (i) Calculate the wavelength of the electromagnetic radiation required to remove an electron from the second shell of a hydrogen atom. **2**
 - (ii) Outline atomic spectral features that cannot be completely explained by the Rutherford-Bohr model. **3**
- (b) In your study of Quanta to Quarks you have performed an investigation to observe radiation emitted from a nucleus using a Wilson Cloud Chamber or a similar detection device.
- (i) Describe how you carried out your investigation. **2**
 - (ii) Isotopes can be used in medicine, agriculture and engineering. **4**
- Choose TWO of these areas and nominate a different isotope for use in each area.
- For each isotope, explain how the properties of the isotope are related to its use.
- (c) Analyse how de Broglie's proposal and supporting experimental evidence led to the move from classical physics to quantum physics. **7**

Question 31 continues on page 36

Question 31 (continued)

- (d) (i) The graph shows the force between nucleons as a function of the distance between them. 2

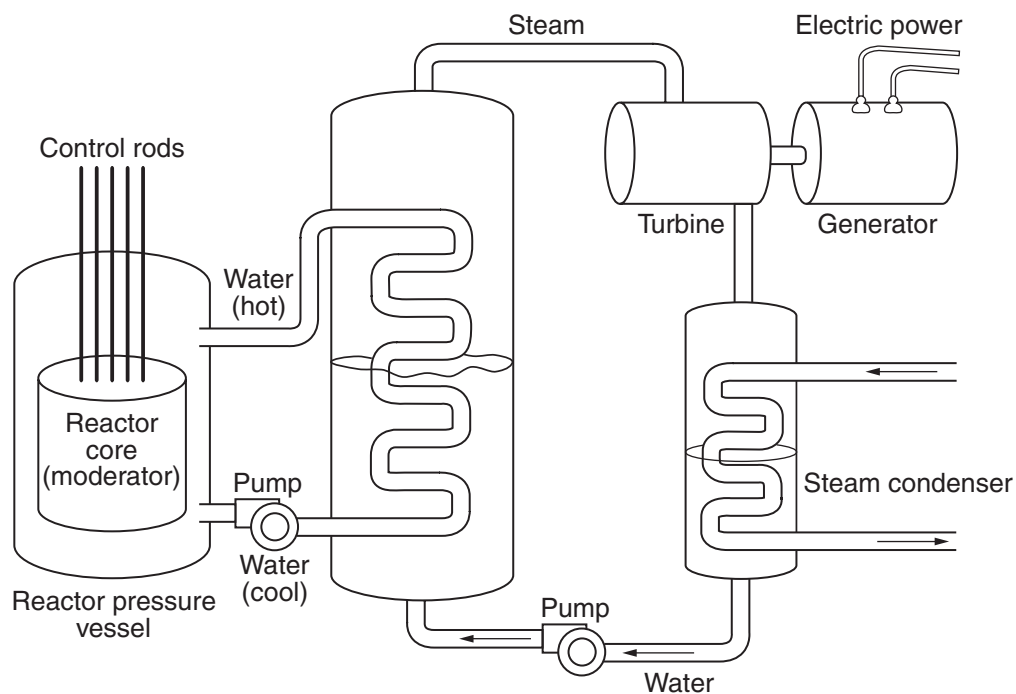


Explain what would happen if two nucleons were separated by the distance indicated as Z on the graph.

Question 31 continues on page 37

Question 31 (continued)

The figure shows the basic features of a nuclear fission reactor.



- (ii) Which part of a nuclear reactor regulates the rate of fission reaction taking place? Justify your answer. **2**
- (iii) Neutron beams can be extracted from nuclear reactors. **3**

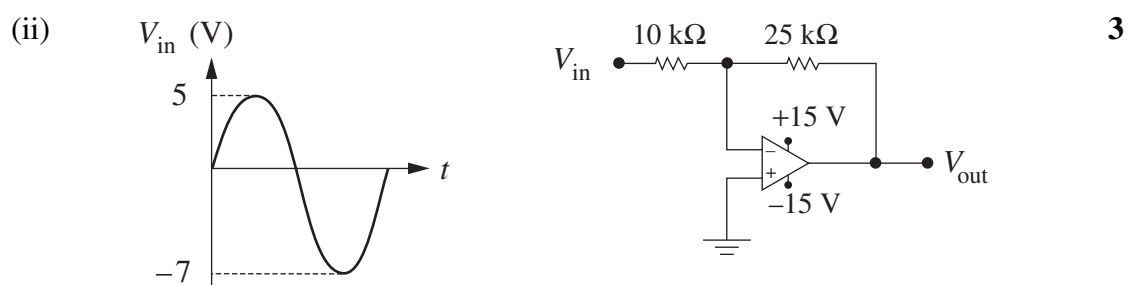
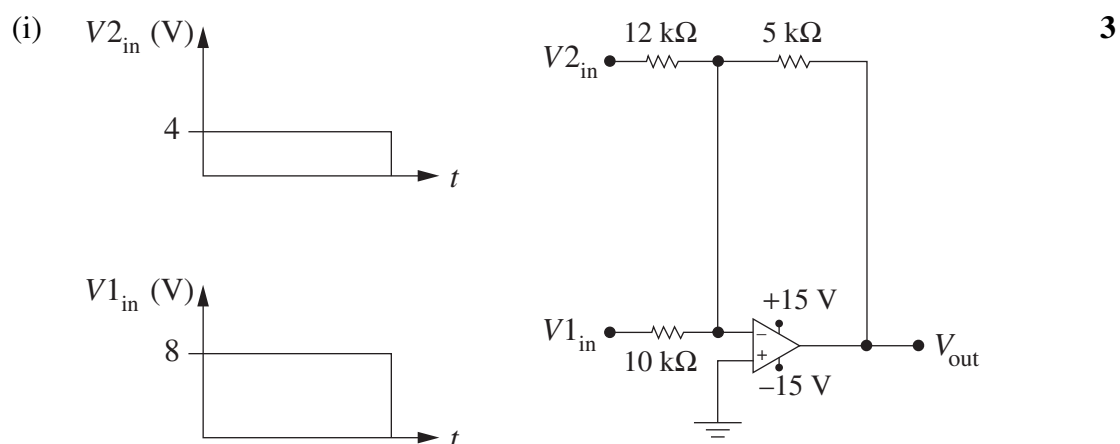
Describe how the neutrons can be used as a probe for investigating the properties of matter.

End of Question 31

Question 32 — The Age of Silicon (25 marks)

- (a) Consider the logic expression $C = (\bar{A} + \bar{B})(A + B)$.
- (i) Using standard symbols, draw a labelled logic diagram in your writing booklet corresponding to this expression. 2
- (ii) Develop a truth table for C . 3
- (b) In your study of the Age of Silicon you have investigated operational amplifiers. The two operational amplifier circuits shown below have the indicated waveforms applied at their inputs.

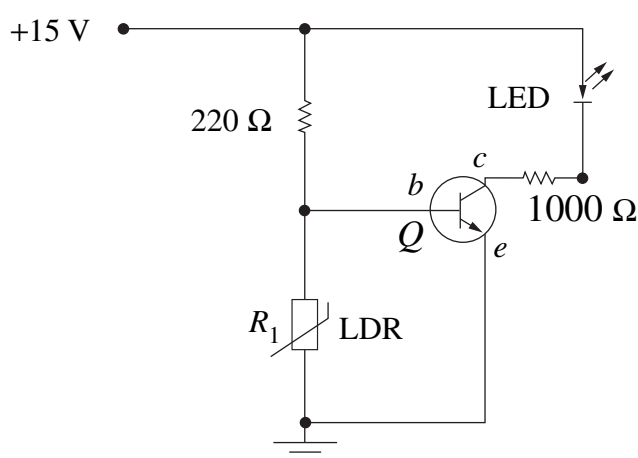
In each case, copy the input waveforms into your writing booklet, and on the same axes sketch the output produced by the circuit. Show all relevant calculations.



Question 32 continues on page 39

Question 32 (continued)

- (c) Analyse how developments in the semiconductor industry could help to reduce energy consumption. 7
- (d) (i) Electronic circuits may use both input and output transducers. Explain the difference between these types of transducer, using an example of each. 3
- (ii) The circuit below is used to illuminate an alarm clock display. 2



R_1 is a light-dependent resistor (LDR) and Q is a transistor with b , c and e as its base, collector and emitter terminals, respectively.

It is found that when the voltage at the base (ie across R_1) is below 0.7 V, the transistor is OFF, preventing current from flowing in the LED. If the base voltage is 0.7 V or more, the transistor is ON, causing 10 mA to flow through the LED.

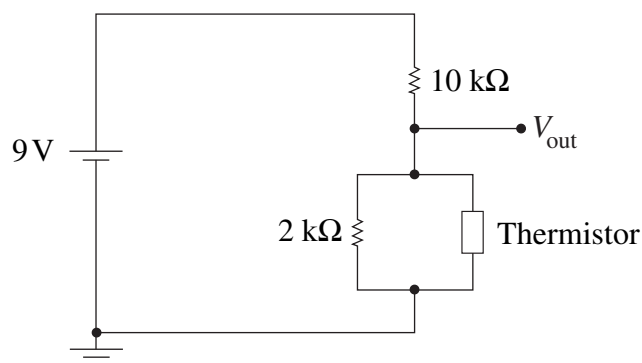
Calculate the value of R_1 at the instant when the transistor switches on.

Question 32 continues on page 40

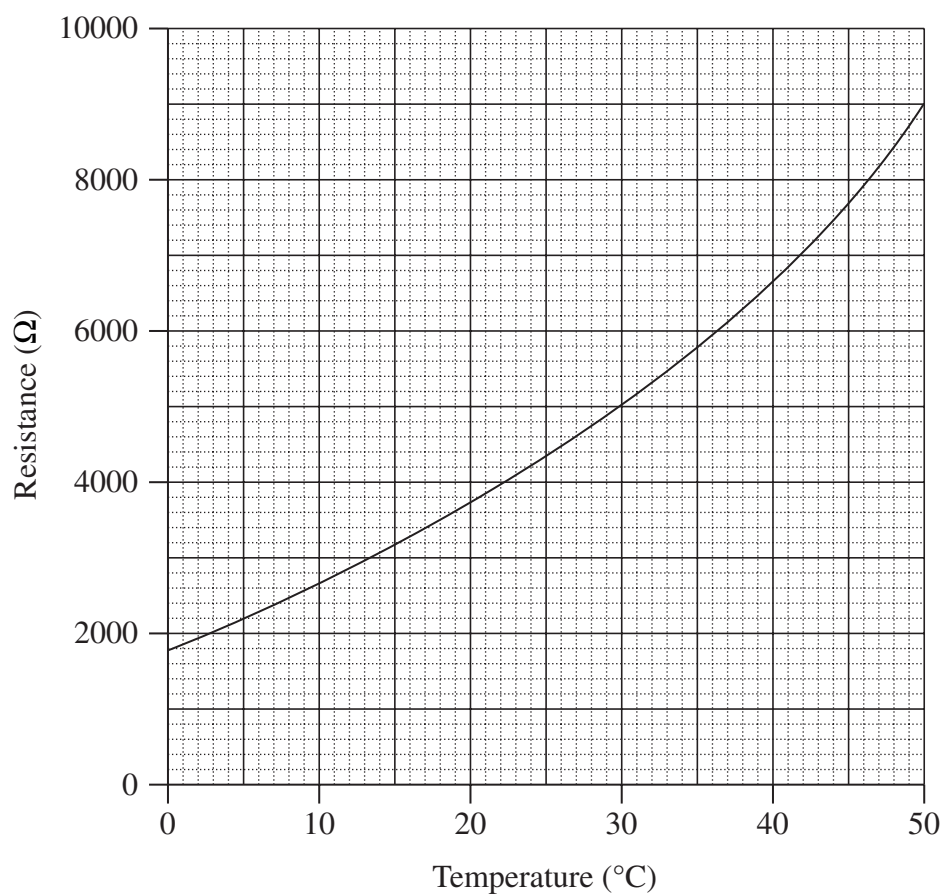
Question 32 (continued)

- (iii) A thermistor is used to construct a thermometer circuit as shown.

2



The thermistor's resistance as a function of temperature is plotted below.



At a certain temperature the output voltage V_{out} is found to be 1.2 V.

What is the temperature of the environment in which the thermometer circuit is located?

End of paper

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DATA SHEET

Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi} \right)$	$2.0 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

FORMULAE SHEET

$$v = f\lambda$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$E = \frac{F}{q}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v_{\text{av}} = \frac{\Delta r}{\Delta t}$$

$$a_{\text{av}} = \frac{\Delta v}{\Delta t} \text{ therefore } a_{\text{av}} = \frac{v - u}{t}$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$W = Fs$$

$$p = mv$$

$$\text{Impulse} = Ft$$

$$E_p = -G \frac{m_1 m_2}{r}$$

$$F = mg$$

$$v_x^2 = u_x^2$$

$$v = u + at$$

$$v_y^2 = u_y^2 + 2a_y \Delta y$$

$$\Delta x = u_x t$$

$$\Delta y = u_y t + \frac{1}{2}a_y t^2$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$F = \frac{Gm_1 m_2}{d^2}$$

$$E = mc^2$$

$$l_v = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{p}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100^{(m_B - m_A)/5}$$

$$\tau = nBIA \cos \theta$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$c = f\lambda$$

$$A_0 = \frac{V_{\text{out}}}{V_{\text{in}}}$$

$$Z = \rho v$$

$$\frac{V_{\text{out}}}{V_{\text{in}}} = - \frac{R_{\text{f}}}{R_{\text{i}}}$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen		KEY										2 He 4.003 Helium																							
3 Li 6.941 Lithium		4 Be 9.012 Beryllium		Atomic Number		79 Au 197.0 Gold		Symbol of element		5 B 10.81 Boron		6 C 12.01 Carbon		7 N 14.01 Nitrogen		8 O 16.00 Oxygen		9 F 19.00 Fluorine		10 Ne 20.18 Neon															
		12 Mg 24.31 Magnesium		Atomic Weight		Gold		Name of element		Aluminium		Silicon		Phosphorus		Sulfur		Chlorine		Argon															
11 Na 22.99 Sodium																																			
19 K 39.10 Potassium		20 Ca 40.08 Calcium		21 Sc 44.96 Scandium		22 Ti 47.87 Titanium		23 V 50.94 Vanadium		24 Cr 52.00 Chromium		25 Mn 54.94 Manganese		26 Fe 55.85 Iron		27 Co 58.93 Cobalt		28 Ni 58.69 Nickel		29 Cu 63.55 Copper		30 Zn 65.41 Zinc		31 Ga 69.72 Gallium		32 Ge 72.64 Germanium		33 As 74.92 Arsenic		34 Se 78.96 Selenium		35 Br 79.90 Bromine		36 Kr 83.80 Krypton	
37 Rb 85.47 Rubidium		38 Sr 87.62 Strontium		39 Y 88.91 Yttrium		40 Zr 91.22 Zirconium		41 Nb 92.91 Niobium		42 Mo 95.94 Molybdenum		43 Tc [97.91] Technetium		44 Ru 101.1 Ruthenium		45 Rh 102.9 Rhodium		46 Pd 106.4 Palladium		47 Ag 107.9 Silver		48 Cd 112.4 Cadmium		49 In 114.8 Indium		50 Sn 118.7 Tin		51 Sb 121.8 Antimony		52 Te 127.6 Tellurium		53 I 126.9 Iodine		54 Xe 131.3 Xenon	
55 Cs 132.9 Caesium		56 Ba 137.3 Barium		57–71 Lanthanoids		72 Hf 178.5 Hafnium		73 Ta 180.9 Tantalum		74 W 183.8 Tungsten		75 Re 186.2 Rhenium		76 Os 190.2 Osmium		77 Ir 192.2 Iridium		78 Pt 195.1 Platinum		79 Au 197.0 Gold		80 Hg 200.6 Mercury		81 Tl 204.4 Thallium		82 Pb 207.2 Lead		83 Bi 209.0 Bismuth		84 Po [209.0] Polonium		85 At [210.0] Astatine		86 Rn [222.0] Radon	
87 Fr [223] Francium		88 Ra [226] Radium		Actinoids		Rutherfordium		Dubnium		Seaborgium		Bohrium		Hassium		Meitnerium		Darmstadtium		Roentgenium															

Lanthanoids

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm [145] Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.0 Ytterbium	71 Lu 175.0 Lutetium
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Actinoids

89 Ac [227] Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np [237] Neptunium	94 Pu [244] Plutonium	95 Am [243] Americium	96 Cm [247] Curium	97 Bk [247] Berkelium	98 Cf [251] Californium	99 Es [252] Einsteinium	100 Fm [257] Fermium	101 Md [258] Mendelevium	102 No [259] Nobelium	103 Lr [262] Lawrencium
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For elements that have no stable or long-lived nuclides, the mass number of the nuclide with the longest confirmed half-life is listed between square brackets.
The International Union of Pure and Applied Chemistry Periodic Table of the Elements (October 2005 version) is the principal source of data. Some data may have been modified.

**2007 HSC Notes from
the Marking Centre
Physics**

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2007 HSC NOTES FROM THE MARKING CENTRE

PHYSICS

Introduction

This document has been produced for the teachers and candidates of the Stage 6 course in Physics. It contains comments on candidate responses to the 2007 Higher School Certificate examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

This document should be read along with the relevant syllabus, the 2007 Higher School Certificate examination, the marking guidelines and other support documents which have been developed by the Board of Studies to assist in the teaching and learning of Physics.

General Comments

In 2007, approximately 9120 candidates attempted the Physics examination. The most popular options were From Quanta to Quarks (45%) and Medical Physics (27%).

Teachers and candidates should be aware that examiners may write questions that address the syllabus outcomes in a manner that requires candidates to respond by integrating their knowledge, understanding and skills developed through studying the course, including the Prescribed Focus Areas. This reflects the fact that the knowledge, understanding and skills developed through the study of discrete sections should accumulate to a more comprehensive understanding than may be described in each section separately. It is important to understand that the Preliminary HSC course is assumed knowledge for the HSC course.

Teachers and candidates are reminded that mandatory skills content in Module 9.1 is examinable in both the Core and Option questions.

Candidates need to be reminded that the answer space provided and the marks allocated are guides to the maximum length of response required. Candidates should use examination time to analyse the question and plan responses carefully, working within that framework to produce clear and concise responses. Responses may include the use of dot points, diagrams and/or tables, and should avoid internal contradictions. This is particularly so in holistic questions which need to be logical and well structured. There was evidence that some candidates had a very poor knowledge of basic definitions specific to terminology associated with the course.

Better responses indicate that candidates had followed the instructions provided on the examination paper. In these responses, candidates:

- set out all working for numerical questions
- thought carefully about the units to be used and the quantities to be substituted into formulae

- did not repeat the question as part of the response
- looked at the structure of the whole question and noted that in some questions the parts follow from each other, ie responses in part (a) lead to the required response in part (b) etc
- used appropriate equipment, for example, pencils and a ruler to draw diagrams and graphs. (A clear plastic ruler helps candidates to plot points that are further from the axes and rule straight lines of best fit.)

In Section II the option question is divided into a number of parts. Candidates should clearly label each part of the question when writing in their answer booklets. In part (c) of the 2007 option questions the best responses presented ideas coherently and included the correct use of scientific principles and ideas. Many candidates wrote a lot of information that was not relevant to the question. Some responses showed evidence of rote learning of an anticipated answer based on a single source. These responses did not address the syllabus content and/or outcomes being assessed and hence did not score full marks. Candidates are required to attempt one question only in Section II, but some candidates responded to more than one option question. Candidates are strongly advised to answer the option they have studied in class.

Section I – Core

Part A – Multiple choice

Question	Correct Response
1	A
2	A
3	B
4	D
5	D
6	D
7	C
8	D

Question	Correct Response
9	C
10	A
11	D
12	A
13	C
14	C
15	A

Part B

Question 16

- (a) Better responses utilised a curved line of best fit as required in the question.
- (b)(ii) Better responses used the correct height when calculating the time of fall before calculating the ball's horizontal velocity. Better responses converted data to SI units when substituting into formulae.
- (c) Better responses used the relevant equation, correctly substituted data, and identified the directions of the vector quantities involved.

Question 17

- (a) Better responses connected the direction of the Earth's rotational velocity with the launch direction and not to the Earth's orbital velocity.
- (b) The best responses were those in which candidates substituted all terms into the correct formula to calculate the satellite's height above the surface. Weaker responses showed some confusion about the satellite described as being on the Earth's surface yet being expected to calculate its height when in orbit.

Question 18

- (a) Better responses provided supporting evidence for the link between the constancy of the speed of light and time, for example time dilation.
- (b) In better responses, candidates correctly substituted into the correct equation.
- (c) Best responses correctly rearranged the relevant equation to calculate the velocity.

Question 19

In better responses, candidates stated Einstein's theory as a hypothesis and then linked it to the parts of the Scientific Method by indicating some or all of the testable predictions that arose from it.

Question 20

In better responses, the candidates clearly expressed a judgement on the effect that the AC generators had on both today's society and the environment.

Question 21

- (a) In better responses, candidates justified the choice of point by identifying the direction of the torque on the coil.
- (b) Better responses identified the correct distance of the line of action of the force.

Question 22

The best responses clearly compared and differentiated between similar structural and operational features of both thermionic and solid state devices.

Question 23

- (c) Better responses provided a clear sequence of cause and effect in the explanation.

Question 24

- (a) Better responses indicated a clear understanding of the nature of the electric field.
- (b) The best responses included a data list that enabled candidates to select the formula to substitute into.

Question 25

Better responses included accurate, fully labelled diagrams and were explicit in their descriptions of function and comparison.

Question 26

- (a) Better responses indicated the differences between AC and DC in terms of alternating magnetic fields.
- (b) The best responses related the energy losses to the properties and structure of transformers.
- (c) Better responses distinguished between energy and power and used the correct time units in the calculations.

Question 27

- (a) In the better responses, candidates linked quantum theory to problems with classical predictions.
- (b) In the better responses, candidates described the use of UV light and the observed spark at the receiver.
- (c)
 - (i) In the best responses, the candidates used the units from the axes when calculating the gradient.
 - (ii) In the better responses, candidates related the gradients to Planck's constant.

Section II – Options

Question 28 — Geophysics

- (a) (ii) In better responses, candidates correctly manipulated the data and equations provided.
- (b)(ii) In the best responses, candidates clearly identified the cause of palaeomagnetism and used this in their explanation.
- (c) The best responses clearly identified two areas linked to remote sensing satellites.
- (d) (i) In the best responses, candidates compared both the sequence and type of wave evident in each seismogram.
 - (ii) In the best responses, candidates related the use of seismograms to nuclear tests.
 - (iii) The best responses included both the specific purpose together with a detailed description of how the seismic information was utilised in their example.

Question 29 — Medical Physics

- (a) (i) The best responses selected the correct equation, derived a correct fraction being reflected and thus calculated the fraction of the ultrasound transmitted.
 - (ii) The best responses explained the dipolar nature of piezoelectric crystals and the need for an oscillating voltage then related these to the production and detection of ultrasound.
- (b)(i) The best responses identified several statements in the article which were either erroneous or oversimplifications and then explained the nature of MRI imaging.
 - (ii) The best responses identified that the main reason for ultrasound not being used was the difference in acoustic impedance between flesh and skull bone.
Some responses correctly noted that ultrasound imaging of the brain was possible in infants through the fontanel.
- (c) Better responses provided detailed descriptions of each technology, often with labelled diagrams and linked these technologies to specific uses. These responses included support of the assessment of the impact on both individuals and the community separately and provided specific answers as evidence.
- (d) (i) In the best responses, candidates used a table to compare both advantages and disadvantages for both CAT and PET in terms of structure and function and to identify the differences between each technology.
 - (ii) The better responses were those where candidates applied their knowledge of the physics of the imaging techniques to the unfamiliar situation by referring to measurement of bone density.

Question 30 — Astrophysics

- (a) (i) The best responses named detection methods and showed how these are used to deduce the presence of a binary star.
- (ii) Better responses used the correct distance units when substituting into the correct linked formulae.
- (b) (i) The better responses described one of the two relevant mandated first-hand investigations and then related the results to one characteristic of stars.
- (c) The better responses used specific examples of the technologies, their advantages and linked these to our scientific understanding of celestial objects. Weaker responses contained an excessive amount of information related to the topic area being examined but had little relevance to the question being asked.
- (d) (i) The better responses identified the correct star and were able to provide sufficient justification.
- (ii) The best responses showed the basic content of an H-R diagram including relevant labels or scales.
- (iii) The better responses used a table to make the comparison between the two stars after having identified their evolutionary stage.

Question 31 — From Quanta to Quarks

- (a) (i) Better responses identified the correct formula and stated that the electron would be removed from the second shell to a higher shell due to the incoming EM radiation.
- (ii) In the best responses, candidates named the unexplained feature and then gave some information about that feature.
- (b) (i) Better responses provided the main features of the investigation which included the position of the radioactive source and an observation of the radiation.
- (ii) Better responses correctly named the radio-isotope and clearly showed the link between its radioactive properties and its use.
- (c) In the best responses, candidates demonstrated a thorough understanding of the move from classical physics to quantum physics linking the relevance of de Broglie's explanations on the limitations of the Bohr atom and the experimental evidence that supported this explanation. The better responses showed clear planning and linkages within the structure of the response whereas weaker responses consisted of an excessive amount of information related to the topic area being examined without addressing the question.
- (d) (i) In the better responses, candidates clearly explained their interpretation of the graph.
- (ii) In the best responses candidates identified and provided a reason for the choice of the relevant component.

Question 32 — The Age of Silicon

- (a) (i) Better responses drew a logic diagram that was labelled clearly and that corresponded to the logic expression given.
- (ii) Better responses developed a truth table for which there were a number of columns, rather than providing the initial inputs and the final output column. In doing so the candidates demonstrated clear understanding of the effect of each gate on an input for the logic diagram drawn in part (i).
- (b) (i) Better responses used a step-by-step circuit analysis when working through the problem. The final result was then shown to be the sum of the individual results for the two input voltages.
- (ii) Better responses analysed the effect of the OP amp on each of the two sections of the input waveform and then showed the resultant waveform.
- (c) Better responses clearly identified links between developments in the semiconductor industry and a specific resultant reduction in energy consumption. Weaker responses often described development in the semiconductor industry without making the link to reducing energy consumption.
- (d) (i) Some better responses provided data in a table to answer the question.
- (ii) Better responses showed all parts of the circuit analysis used to determine R1.
- (iii) Better responses showed all parts of the circuit analysis and identified that the thermistor was in parallel with the 1k resistor.

Physics

2007 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
Section I			
Part A			
1	1	9.2.2.2.8, 9.2.3.3.4	H9
2	1	9.2.4.2.9, 14.1(f)	H6, H14
3	1	9.2.1.2.3, 12.4(b)	H7, H12
4	1	9.2.3.3.2, 12.4(b), 14.1(f)	H9, H12, H14
5	1	9.2.2.3.1, 14.1(a)	H6, H14
6	1	9.3.1.1.6, 12.4(b), 14.1(f)	H9, H12, H14
7	1	9.3.2.2.3, 9.3.2.2.4, 14.1(a)	H9, H14
8	1	9.3.2.3.2, 14.1(f)	H9, H14
9	1	9.3.2.3.4, 14.1(f)	H9, H14
10	1	9.4.2.3.2, 9.4.2.2.5	H7, H10
11	1	9.4.1.3.3, 12.4(b)	H9, H12
12	1	9.4.4.2.1	H8, H10
13	1	9.4.1.3.3, 9.4.1.2.3, 12.4(b), 14.1(f)	H9, H12, H14
14	1	9.4.3.2.7	H10
15	1	9.3.2.2.5, 9.3.4.2.4, 9.4.2.2.6, 9.2.1.2.2.	H6, H7, H9, H10
Section II			
Part B			
16 (a)	2	9.2.2.2.1, 9.2.2.3.2, 13.1(f)	H6, H13
16 (b) (i)	1	9.2.2.2.1, 9.2.2.3.1, 9.2.2.3.2, 12.3(b)	H6, H12
16 (b) (ii)	2	9.2.2.2.1, 9.2.2.3.2, 9.2.2.3.1, 12.4(b)	H6, H12
17 (a)	1	9.2.2.2.6, 13.1(e)	H6, H13
17 (b)	3	9.2.2.2.9, 9.2.2.3.5, 12.4(b)	H6, H12
18 (a)	2	9.2.4.2.7	H6
18 (b)	2	9.2.3.3.5, 12.4(b)	H6, H7, H12

Question	Marks	Content	Syllabus outcomes
18 (c)	3	9.2.4.3.5, 12.4(b)	H6, H12
19	6	9.2.4.3.4, 14.1(e-h), 14.3(b)	H2, H6, H14
20	4	9.3.3.2.1, 9.3.3.2.5	H4, H7
21 (a)	1	9.3.1.2.3, 9.3.1.2.4, 9.3.1.3.4	H9
21 (b)	2	9.3.1.2.3, 9.3.1.3.4, 12.4(b)	H9, H12
21 (c)	2	9.3.1.2.3, 9.3.1.3.4, 12.4(b)	H9, H12
22	4	9.4.3.2.8	H10
23 (a)	1	9.4.4.2.7, 14.1(a)	H10, H14
23 (b)	1	9.4.4.2.5	H10
23 (c)	2	9.4.4.2.7, 9.4.4.3.3	H9
24 (a)	1	9.4.1.2.4, 8.3.2.2.1, 13.1(e)	H9, H13
24 (b)	2	9.4.1.3.3, 12.4(b)	H9, H12
25	4	9.3.3.2.2, 14.3(e)	H7, H9, H14
26 (a)	2	9.3.2.2.4, 9.3.3.32	H9
26 (b)	2	9.3.4.3.3	H7
26 (c)	2	9.3.4.2.4, 8.3.4.3.2, 12.4(b)	H7, H12
27 (a)	3	9.4.2.2.5, 9.4.2.2.6	H10
27 (b)	2	9.4.2.2.1	H10
27 (c) (i)	2	9.4.2.2.6, 14.3(c)	H10, H14
27 (c) (ii)	1	9.4.2.2.6	H10

Section II

Question 28 — Geo Physics

28 (a) (i)	3	9.5.1, 9.5.1.2.1, 9.5.3.2.1, 9.5.3.2.2, 9.5.3.2.5, 14.1(a), 13.1(f)	H7, H8, H13, H14
28 (a) (ii)	2	9.5.1, 9.5.1.2.1, 9.5.3.2.1, 9.5.3.2.2, 13.1(d), 14.1(f)	H7, H8, H13, H14
28 (b) (i)	2	9.5.4.3.1, 13.1(e)	H9, H13
28 (b) (ii)	4	9.5.4.2.2, 9.5.4.2.3, 14.1(a)	H9, H14
28 (c)	7	9.5.2.2.1, 9.5.2.2.2, 9.5.2.2.3, 9.5.2.2.4, 9.5.2.2.5, 9.5.2.3.1, 14.3(b)	H1, H3, H6, H9, H14

Question	Marks	Content	Syllabus outcomes
28 (d) (i)	3	9.5.3.2.1, 12.3(c), 14.1(a)	H7, H8, H12, H14
28 (d) (ii)	2	9.5.5.2.2, 9.5.5.1	H7, H8
28 (d) (iii)	2	9.5.5.2.1, 9.5.5.2.2, 9.5.5.1	H7, H8
Section II			
Question 29 — Medical Physics			
29 (a) (i)	2	9.6.1.2.5, 9.6.1.3.5, 12.4(b), 13.1(d)	H8, H12, H13
29 (a) (ii)	3	9.6.1.2.2	H7, H8
29 (b) (i)	4	9.6.4.3.4, 12.3(c), 12.4(f), 14.1(h)	H8, H9, H10, H12, 14
29 (b) (ii)	2	9.6.1.3.1	H8
29 (c)	7	9.6.4.3.5, 14.3(b)	H4, H9, H10, H14
29 (d) (i)	3	9.6.4.3.4, 12.3(c)	H8, H12
29 (d) (ii)	4	9.6.4.2.2, 9.6.4.2.3, 9.6.4.2.4, 9.6.4.3.2, 14.1(a)	H10, H14
Section II			
Question 30 — Astrophysics			
30 (a) (i)	2	9.7.5.2.1	H6, H9
30 (a) (ii)	3	9.7.4.3.1, 8.2.3.2.4, 12.4(b), 13.1(d)	H8, H12, H13
30 (b) (i)	2	9.7.3.3.1, 14.1(a)	H7, H8, H14
30 (b) (ii)	4	9.7.3.2.5, 9.7.5.2.1	H7, H8
30 (c)	7	9.7.1.2.5, 9.7.2.3.2, 9.7.4.2.5, 14.3(b)	H1, H3, H10, H14
30 (d) (i)	1	9.7.4.2.1, 9.7.4.3.1, 12.3(c)	H7, H8, H12
30 (d) (ii)	3	9.7.6.3.1, 9.7.6.3.3, 12.3(c), 13.1(f), 14.1(g)	H8, H12, H13, H14
30 (d) (iii)	3	9.7.6.3.2	H7, H8, H9
Section II			
Question 31 — From Quanta to Quarks			
31 (a) (i)	2	9.8.1.2.5, 9.8.1.3.3, 12.4(b), 13.1(d)	H8, H12, H13
31 (a) (ii)	3	9.8.1.3.4	H7, H8
31 (b) (i)	2	9.8.3.3.1, 9.8.3.2.1	H10
31 (b) (ii)	4	9.8.4.2.2, 9.8.4.3.2	H10
31 (c)	7	9.8.2, 9.8.2.2.1, 9.8.2.2.4, 14.3(b)	H1, H8, H14

Question	Marks	Content	Syllabus outcomes
31 (d) (i)	2	9.8.3.2.8, 12.3(c), 14.1(a)	H10, H12, H14
31 (d) (ii)	2	9.8.3.2.11, 9.8.4.2.1, 12.3(c)	H7, H10, H12
31 (d) (iii)	3	9.8.4.2.3	H7, H10
Section II			
Question 32 — The Age of Silicon			
32 (a) (i)	2	9.9.5.2.1, 9.9.5.3.2, 13.1(d)	H7, H13
32 (a) (ii)	3	9.9.5.2.1, 9.9.5.3.2, 14.1(f)	H7, H14
32 (b) (i)	3	9.9.6.2.2, 9.9.6.3.3, 12.3(c), 14.1(f), 12.4(b)	H7, H12, H14
32 (b) (ii)	3	9.9.6.2.2, 9.9.6.2.8, 9.9.6.3.1, 12.3(c), 12.4(b), 14.1(f)	H7, H12, H14
32 (c)	7	9.9.4.2.3, 9.9.4.3.3, 14.3(b)	H4, H7, H14
32 (d) (i)	3	9.9.3.2.1, 9.9.3.3.3	H7
32 (d) (ii)	2	9.9.3.3.2, 9.9.2.2.4, 12.4(b)	H7, H12
32 (d) (iii)	2	9.9.3.3.2, 9.9.3.2.6, 9.9.3.2.4, 9.9.3.2.5, 9.9.3.3.1, 12.4(b), 14.1(f)	H7, H12, H14

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Section I, Part B

Question 16 (a)

Outcomes assessed: H6, H13

MARKING GUIDELINES

Criteria	Marks
• Draws line of best fit which is: - a curve - close to all points - has approximately the same number (if any) of points above and below the line AND EITHER • Correctly plots all seven points OR • Plots SIX points correctly and seventh incorrectly by not more than TWO divisions OR • Plots FIVE points correctly and sixth and seventh incorrectly by not more than ONE division	2
• Correctly plots all seven points ONLY OR • Plots SIX points correctly and seventh incorrectly by not more than TWO divisions ONLY OR • Plots FIVE points correctly and sixth and seventh incorrectly by not more than ONE division ONLY OR • Draws a line (close to all points and approximately the same number of points above and below the line)	1

Question 16 (b) (i)

Outcomes assessed: H6, H12, H14

MARKING GUIDELINES

Criteria	Marks
- Predicts the range as $92 \text{ cm} \pm 1 \text{ cm}$ (whether or not marked on the graph) OR - Predicts the range consistent with correct extrapolation of graph	1

Question 16 (b) (ii)

Outcomes assessed: H6, H12

MARKING GUIDELINES

Criteria	Marks
Calculates correct velocity for range chosen	2
Identifies correct equations to calculate velocity	1

Question 17 (a)

Outcomes assessed: H6, H13

MARKING GUIDELINES

Criteria	Marks
- Draws arrow towards East from X - Provides an appropriate justification	1

Question 17 (b)

Outcomes assessed: H6, H12

MARKING GUIDELINES

Criteria	Marks
- Identifies correct formula - Correctly substitutes G, M and T - Subtracts the Earth's radius to calculate value r	3
- Any two of the above OR - Recalls correct value for satellite radius and subtracts radius of Earth	2
- Any one of the above OR - Recalls correct value of altitude	1

Question 18 (a)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
- Refers to c as a constant, so time is relative to observer - Refers to c as a constant, so time can be used to define length OR - States an example of time dilation of a rapidly moving object	2
- Refers to c as a constant, so time is relevant to observer OR - Refers to c as a constant, so that time can be used to define length OR - States an example of time dilation of a rapidly moving object	1

Question 18 (b)

Outcomes assessed: H6, H7, H12

MARKING GUIDELINES

Criteria	Marks
Identifies correct formula AND correctly substitutes	2
Identifies correct formula	1

Question 18 (c)

Outcomes assessed: H6, H12

MARKING GUIDELINES

Criteria	Marks
- Identifies correct formula and correctly substitutes - Manipulates correct formula to extract v	3
Identifies correct formula and correctly substitutes	2
Identifies correct formula	1

Question 19

Outcomes assessed: H2, H6, H14

MARKING GUIDELINES

Criteria	Marks
- Demonstrates a thorough knowledge of scientific method and Einstein's Theory of Special Relativity and the evidence supporting it - Identifies the evidence to support Einstein's Theory of Special Relativity - Outlines the development and acceptance of the theory - Links the development of the theory and the evidence supporting it to the steps in the model of scientific method - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	5-6
- Demonstrates knowledge of scientific method and Einstein's Theory of Special Relativity and the evidence supporting it AND EITHER - Describes the development and acceptance of the theory and evidence used to support it OR - Describes some aspects of the theory, some evidence supporting it and attempts to relate this to the steps in the model of scientific method	3-4
- Demonstrates a limited knowledge of scientific method and Einstein's Theory of Special Relativity AND EITHER - Describes some aspects of Einstein's theory OR - Describes some parts of the model of scientific method including 3 steps	1-2

Question 20

Outcomes assessed: H4, H7

MARKING GUIDELINES

Criteria	Marks
Describes positive and/or negative effects of the development of AC generators on society and environment to justify their value judgment on its effects	3-4
- Provides some positive and/or negative effects of the development of AC generators on society and environment OR - Provides value judgement of its effects	1-2

Question 21 (a)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Gives correct answer with justification	1

Question 21 (b)

Outcomes assessed: H9, H12

MARKING GUIDELINES

Criteria	Marks
- Identifies correct equation - Calculates correct value of torque	2
Identifies correct equation	1

Question 21 (c)

Outcomes assessed: H9, H12

MARKING GUIDELINES

Criteria	Marks
- Identifies correct equation - Manipulates to calculate the number of turns	2
Identifies correct equation	1

Question 22

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
- Provides reasons why solid state devices have replaced thermionic devices - Shows how solid state and thermionic devices are similar /different	3–4
Sketches in general terms the features, solid state and thermionic devices	2
Sketches in general terms the function of either device OR	1
Identifies that the two devices perform a similar function	

Question 23 (a)

Outcomes assessed: H10, H14

MARKING GUIDELINES

Criteria	Marks
Recognises and names what scientists are trying to achieve	1

Question 23 (b)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
Identifies one property and the effect that critical temperature has on it	1

Question 23 (c)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
States there is a force of repulsion between the magnet and the superconductor and shows how the repulsion occurs	2
States there is a force of repulsion between the magnet and the superconductor	1

Question 24 (a)

Outcomes assessed: H9, H13

MARKING GUIDELINES

Criteria	Marks
Draws lines with direction arrows, which are symmetrical L/R AND point in the correct direction	1

Question 24 (b)

Outcomes assessed: H9, H12

MARKING GUIDELINES

Criteria	Marks
- Correctly identifies formulae - Substitutes given values correctly	2
Correctly identifies formula $F = Eq$	1

Question 25

Outcomes assessed: H7, H9, H14

MARKING GUIDELINES

Criteria	Marks
- Using identified common components of both, explains how motors and generators work with reference to a comparison of energy inputs and outputs - Includes a diagram showing at least TWO features	4
- Using identified common components of both, describes differences in function between motors and generators - Includes a diagram showing at least TWO features	3
- Describes how a motor and a generator works OR - Identifies common features - Includes a diagram showing at least TWO features	2
Lists common features	1

Question 26 (a)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Explains the need for AC current in terms of change in magnetic flux	2
- States that transformers will not work with DC current OR - Explains the need for AC current with no reference to magnetic flux	1

Question 26 (b)

Outcomes assessed: H7

MARKING GUIDELINES

Criteria	Marks
Sketches in general terms TWO possible causes of energy loss	2
Sketches in general terms ONE possible cause of energy loss	1

Question 26 (c)

Outcomes assessed: H7, H12

MARKING GUIDELINES

Criteria	Marks
- Correctly states the input and output power. - Calculates the power loss and the energy lost in 8 hours	2
- Correctly calculates the input OR output power OR - States that $E = Pt$ or VIt	1

Question 27 (a)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
- Describes a black body in terms of its features and radiation curve - Links the shape of the curve to emission of energy in packets/quanta	3
- Describes a feature of a black body or radiation curve - States the emitted energy is quantized	2
Any of the above	1

Question 27 (b)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
• Provides features of the photoelectric effect observed by Hertz	2
• Provides limited features of the observed photoelectric effect	1

Question 27 (c) (i)

Outcomes assessed: H10, H14

MARKING GUIDELINES

Criteria	Marks
• Correctly calculates the gradient of sodium	2
• Attempts to calculate the gradient of sodium	1

Question 27 (c) (ii)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
• Identifies the significance of the gradient	1

Question 28 (a) (i)

Outcomes assessed: H7, H8, H13, H14

MARKING GUIDELINES

Criteria	Marks
• Provides features and characteristics of the differences in rigidity and density of rock in the mantle and core • Makes the connection between the speed of P and S waves at different depths and the differences rigidity and density	3
• Provides features and characteristics of the differences in rigidity and density of rock in the mantle and core • States the speed of P or S waves at different depths	2
• States any observable relationship between a wave type, depth, rigidity or density	1

Question 28 (a) (ii)

Outcomes assessed: H7, H8, H13, H14

MARKING GUIDELINES

Criteria	Marks
Correctly calculates the ratio of the velocities and provides the correct answer	2
States the correct formula that could be used to calculate the ratio of the velocities	1

Question 28 (b) (i)

Outcomes assessed: H9, H13

MARKING GUIDELINES

Criteria	Marks
- Describes a plausible investigation to model a variation in inclination - Provides features and characteristics of the inclination obtained	2
- Sketches in general terms the modeling investigation OR - Correct statement of what magnetic inclination is	1

Question 28 (b) (ii)

Outcomes assessed: H9, H14

MARKING GUIDELINES

Criteria	Marks
- Provides how palaeomagnetism is formed - Provides how the ancient south pole can be determined from each continent - Outline how the lining up of the ancient south pole provides evidence for continental drift	4
Any TWO of the above	3
ONE of the above	2
States that iron in lava aligns with Earth's magnetic field as it cools	1

Question 28 (c)

Outcomes assessed: H1, H3, H6, H9, H14

MARKING GUIDELINES

Criteria	Marks
• Demonstrates a thorough understanding of two areas of knowledge of Earth that the introduction of satellites have contributed to • Identifies areas of knowledge of Earth that were previously inaccessible • Identifies information that satellites can tell us about Earth and that satellites have contributed to our knowledge of Earth • Makes the relationship between the introduction of satellites and how they have contributed to the areas of knowledge of Earth that were previously inaccessible evidence • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
• Demonstrates a sound understanding of the areas of knowledge of Earth that were inaccessible before the introduction of satellites • Describes information that can be determined about Earth from satellites • Describes the areas of knowledge of Earth that satellites have contributed to	4–5
• Demonstrates a basic knowledge of how satellites have contributed to our understanding of Earth AND EITHER • Outlines areas of knowledge that have satellites have contributed to OR • Describes information that can be determined about Earth from Satellites	2–3
• Demonstrates limited knowledge of satellites and our knowledge of Earth AND EITHER • Provides some relevant information about what satellites can tell us about Earth • Provides some relevant information about an area of Knowledge of Earth that was previously inaccessible	1–2

Question 28 (d) (i)

Outcomes assessed: H7, H8, H12, H14

MARKING GUIDELINES

Criteria	Marks
- Shows how the two seismograms are similar/different in terms of both P and S waves - States the cause of the differences between the seismograms in terms of the original disturbance	3
- Makes a point of comparison between the two seismograms - Describes the difference in the types of waves produced by an earthquake and nuclear test	2
Makes a valid comparison between the two seismograms eg. intensity	1

Question 28 (d) (ii)

Outcomes assessed: H7, H8

MARKING GUIDELINES

Criteria	Marks
States TWO pieces of information obtained from seismograms	2
States ONE piece of information obtained from seismograms	1

Question 28 (d) (iii)

Outcomes assessed: H7, H8

MARKING GUIDELINES

Criteria	Marks
- Gives features of seismogram studies in a natural hazard reduction application or mineral exploration OR - Gives ONE feature of seismogram studies in a natural hazard reduction application or mineral exploration and correctly outlines the use of the feature	2
Identifies a specific use for seismogram studies in natural hazard reduction or mineral exploration	1

Question 29 (a) (i)

Outcomes assessed: H8, H12, H13

MARKING GUIDELINES

Criteria	Marks
- Correctly identifies and substitutes data into appropriate equation - Deducts answer	2
Correctly identifies and substitutes data into appropriate equation	1

Question 29 (a) (ii)

Outcomes assessed: H7, H8

MARKING GUIDELINES

Criteria	Marks
- Identifies the structure of a piezoelectric crystal - Shows how it acts as a source of ultrasonic waves - Shows how it acts as a detector of ultrasonic waves OR - Shows a detailed understanding of how it acts as both source and detector	3
- Identifies the structure of a piezoelectric crystal AND shows how it acts as a source of ultrasonic waves OR - Identifies the structure of a piezoelectric crystal AND shows how it works as a detector of ultrasonic waves OR - Shows how it acts as both source and detector	2
- Identifies the structure of a piezoelectric crystal OR - Shows how it acts as a source of ultrasonic waves OR - Shows how it acts as a detector of ultrasonic waves	1

Question 29 (b) (i)

Outcomes assessed: H8, H9, H10, H12, H14

MARKING GUIDELINES

Criteria	Marks
Provides a judgment of THREE statements from the article in terms of their scientific accuracy showing a detailed understanding of the physics involved	4
- Provides a judgment of TWO statements in terms of their scientific accuracy showing a detailed understanding of the physics involved OR - Provides a judgment on THREE statements showing limited understanding of the physics involved	3
- Provides a judgment on ONE statement in term of its scientific accuracy showing a detailed understanding of the physics involved. - Provides a judgment on TWO statements showing limited understanding of the physics involved	2
Attempts to provide a judgment of ONE statement in terms of its scientific accuracy demonstrating limited understanding of the physics involved	1

Question 29 (b) (ii)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
Identifies that the percentage of waves transmitted into the brain is too small for imaging to occur and relates this to the large percentage of waves reflected due to the large difference in acoustic impedance of brain and skull	2
Identifies that the percentage of waves transmitted into the brain is too small for imaging to occur	1

Question 29 (c)

Outcomes assessed: H4, H9, H10, H14

MARKING GUIDELINES

Criteria	Marks
• Demonstrates a thorough understanding of production and use of x-rays, CAT scans and MRI • Describe the effect of these technologies on lives of individuals and the wider community • Makes a judgment of the effect of these technologies on lives of individuals and the wider community • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
• Demonstrates a sound understanding of production and use of x-rays, CAT scans and MRI • Outlines the effect of these technologies on the lives of individuals and/or the wider community	4–5
• Demonstrates a basic knowledge of production and/or use of x-rays, CAT scans or MRI AND EITHER • Outlines the effect of these technologies on the lives of individuals and the wider community OR • Describes the effect of a technology on the lives of individuals OR the wider community	2–3
• Demonstrates limited knowledge of production and/or use of x-rays and/or CAT scans and/or MRI OR • Outlines the effect of a technology on the lives of individuals or the wider community	1

Question 29 (d) (i)

Outcomes assessed: H8, H12

MARKING GUIDELINES

Criteria	Marks
• Provides a detailed comparison of the advantages and disadvantages of CAT and PET scans with reference to both structure AND function of the brain	3
• Compares some advantages and disadvantages of CAT and PET scans with reference to EITHER structure or function of the brain	2
• Describes a PET scan OR • Describes a CAT scan	1

Question 29 (d) (ii)

Outcomes assessed: H10, H14

MARKING GUIDELINES

Criteria	Marks
- Identifies the imaging techniques and provides a detailed description of each - Provides a statement of comparison	4
- Identifies the imaging techniques - Provides a detailed description of each	3
- Identifies imaging techniques and provides a limited description of each OR - Identifies one imaging technique and provides a detailed description of it	2
- Identifies imaging techniques OR - Identifies an imaging technique and provides a limited description of it	1

Question 30 (a) (i)

Outcomes assessed: H6, H9

MARKING GUIDELINES

Criteria	Marks
Provides features of how the existence of binary star may be deduced by referencing BOTH the orbital motion AND the companion	2
Provides features of how the existence of binary star may be deduced by referencing EITHER the orbital motion OR the companion	1

Question 30 (a) (ii)

Outcomes assessed: H8, H12, H13

MARKING GUIDELINES

Criteria	Marks
Correctly calculates apparent magnitude	3
Correctly substitutes into appropriate formulae	2
- Uses the inverse square law to identify a reduction in intensity by a factor of 100 OR - Identifies one of the correct formula	1

Question 30 (b) (i)

Outcomes assessed: H7, H8, H14

MARKING GUIDELINES

Criteria	Marks
• Provides features of the investigation and results obtained	2
• Provides features of either the investigation OR results obtained OR • Outlines the investigation and results obtained	1

Question 30 (b) (ii)

Outcomes assessed: H7, H8

MARKING GUIDELINES

Criteria	Marks
• Show how properties of a star are both deduced	4
• Lists TWO properties of a star and explains how one of these is deduced	3
• Lists TWO properties of a star OR • Lists a property and explains how it is deduced	2
• Lists a property of a star	1

Question 30 (c)

Outcomes assessed: H1, H3, H10, H14

MARKING GUIDELINES

Criteria	Marks
• Demonstrates a thorough understanding of technologies used in modern astrophysics that have changed our understanding of celestial objects • Describes technologies used in modern astrophysics • Describes how each technology has changed our scientific understanding of celestial bodies • Makes a judgment on the ways technologies used in modern astrophysics have changed our understanding of celestial bodies • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
• Demonstrates a sound understanding of technology used in modern astrophysics that have changed our scientific understanding of celestial bodies • Describes a technology used in modern astrophysics • Describes how this technology has changed our understanding of celestial bodies	4–5
• Demonstrates a basic knowledge of technology used in astrophysics AND EITHER • Describes a technology used in astrophysics OR • Describes how technology has changed our scientific understanding	2–3
• Demonstrates limited knowledge of technology used in astrophysics • Provides some relevant information about technology used in astrophysics or how our scientific knowledge has changed about celestial bodies	1

Question 30 (d) (i)

Outcomes assessed: H7, H8, H12

MARKING GUIDELINES

Criteria	Marks
• Identifies the most distant star AND provides a correct reason	1

Question 30 (d) (ii)

Outcomes assessed: H8, H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
• Draws HR diagram with main sequence and plots positions of the 5 stars correctly	3
• Draws an appropriate HR diagram with main sequence	2
• Draws an appropriate HR diagram	1

Question 30 (d) (iii)

Outcomes assessed: H7, H8, H9

MARKING GUIDELINES

Criteria	Marks
• Shows similarities or differences of one physical property, nuclear reactions and evolutionary stage	3
• Shows similarities or differences of ONE physical property, nuclear reactions and evolutionary stage OR • Shows similarities or differences of ONE physical property, nuclear reactions and evolutionary stage	2
• Shows similarities or differences of a physical property or nuclear reactions or evolutionary stage	1

Question 31 (a) (i)

Outcomes assessed: H8, H12, H13

MARKING GUIDELINES

Criteria	Marks
• Correctly substitutes values into correctly identified equation	2
• Identifies correct equation	1

Question 31 (a) (ii)

Outcomes assessed: H7, H8

MARKING GUIDELINES

Criteria	Marks
• Outlines more than TWO spectral features	3
• Outlines TWO spectral features OR • Names more than TWO spectral features	2
• Outlines a spectral feature OR • Names TWO spectral features	1

Question 31 (b) (i)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
• Provides features of the investigation	2
• Outlines the investigation OR • Provides ONE feature of the investigation	1

Question 31 (b) (ii)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
• Selects two different isotopes, each relevant to the nominated field • Shows how the properties of each are related to their respective uses	3–4
• Correctly chooses an isotope relevant to a nominated field and shows how the properties are related to its use OR • Selects two different isotopes each relevant to the nominated field but with little or no explanation	1–2

Question 31 (c)

Outcomes assessed: H1, H8, H14

MARKING GUIDELINES

Criteria	Marks
• Demonstrates a thorough understanding that the limitations of classical physics gave birth to quantum physics • Identifies de Broglie's concepts/ideas and supporting experimental evidence caused the move • Describes the evidence and ideas/concepts • Makes the relationship evident between the cause of the move from classical physics to quantum physics and de Broglie's ideas and supporting evidence • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
• Demonstrates a sound understanding that the limitations of classical physics gave birth to quantum physics • Describes de Broglie's concepts/ideas and supporting experimental evidence that caused the move • Outlines the limitations of classical physics OR what quantum physics attempts to explain	4–5
• Demonstrates a basic knowledge that the limitations of classical physics gave birth to quantum physics AND EITHER • Outlines the limitations of classical physics or what quantum physics attempts to explain OR • Outlines de Broglie's ideas/developments OR supporting experimental evidence	2–3
• Demonstrates limited knowledge that the limitations of classical physics gave birth to quantum physics AND EITHER • Provides some relevant information about de Broglie's ideas/concepts or supporting experimental evidence OR • Provides some relevant information about the limitations of classical physics or what quantum physics attempts to explain	1

Question 31 (d) (i)

Outcomes assessed: H10, H12, H14

MARKING GUIDELINES

Criteria	Marks
• States what would have happened and provides a reason why	2
• States what would happen without providing a reason	1

Question 31 (d) (ii)

Outcomes assessed: H7, H10, H12

MARKING GUIDELINES

Criteria	Marks
• Identifies 'control rods'	2
• Provides a reason why they regulate the rate of fission reaction	
• Identifies 'control rods'	1
OR • Outlines how control rods work	

Question 31 (d) (iii)

Outcomes assessed: H7, H10

MARKING GUIDELINES

Criteria	Marks
• Provides a thorough link between TWO properties of a neutron and its ability to investigate the properties of the matter	3
OR • Outlines THREE features of neutrons and provides a link with its ability to investigate matter	
• Outlines TWO features of neutron behaviour and provides a sound link between one of these properties and its ability to investigate the properties of matter	2
OR • Outlines THREE features of neutron properties	
• Outlines features of neutron behaviour without providing a link to its application	1
OR • Outlines a feature of a neutron and links its ability to investigate properties of matter	

Question 32 (a) (i)

Outcomes assessed: H7, H13

MARKING GUIDELINES

Criteria	Marks
Uses correct symbols for gates and logic operations and draws correct circuit with inputs and output	2
- Uses incorrect symbols but shows inputs and outputs correctly OR - Uses correct symbols	1

Question 32 (a) (ii)

Outcomes assessed: H7, H14

MARKING GUIDELINES

Criteria	Marks
Provides all correct columns OR four correct columns with a maximum of one logic error	3
- Provides columns 1–4 correctly with a maximum of one logic error OR - Provides columns 1–3 correctly	2
- Has incorrect truth table but gives correct equivalent gate for C OR - Provides 1st and 2nd columns correctly with a maximum of one logic error	1

Question 32 (b) (i)

Outcomes assessed: H7, H12, H14

MARKING GUIDELINES

Criteria	Marks
Correctly draws the output waveform with supporting calculations also provided	3
- Calculates correct gains but does not show correct waveform OR - Uses correct formulae but makes a mistake with V_2 or V_1 gain and then carries this to the final waveform plot	2
- Uses correct formulae but makes mistakes in applying them to the circuit eg does not identify R_1, R_2, and R_f correctly OR - Draws correct waveform without justification or identifies a negative voltage OR - Identifies circuit as a summing amplifier	1

Question 32 (b) (ii)

Outcomes assessed: H7, H12, H14

MARKING GUIDELINES

Criteria	Marks
Correctly draws the output waveform with supporting information also provided	3
- Calculates correct gain but does not show correct waveform for one or other of the half-cycles OR - Makes a numerical mistake and draws the correct waveform based on the incorrect calculations	2
- Identifies correct formula OR - Identifies circuit as an inverting amplifier	1

Question 32 (c)

Outcomes assessed: H4, H7, H14

MARKING GUIDELINES

Criteria	Marks
- Demonstrates a thorough understanding of developments in semiconductors that have led to a reduction of energy consumption - Identifies specific examples of solid state devices that have resulted in energy savings - Identifies how these devices reduce energy consumption - Makes the relationship evident between developments in semiconductor industry and reduction in energy consumption - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
- Demonstrates a sound understanding of developments in semiconductors that have led to a reduction of energy consumption - Describes examples of semiconductors that have resulted in energy savings	4–5
- Demonstrates a basic knowledge of developments in semiconductors that have led to a reduction of energy consumption AND EITHER - Outlines examples of semiconductors that have led to a reduction of energy consumption OR - Describes an example of a semiconductor that has led to a reduction of energy consumption	2–3
- Demonstrates limited knowledge of developments in semiconductors that have led to a reduction of energy consumption - Provides general information of semiconductors that have led to a reduction in energy consumption	1

Question 32 (d) (i)

Outcomes assessed: H7

MARKING GUIDELINES

Criteria	Marks
- Provides an explanation of how input and output transducers are different - Provides an example of an input transducer and an output transducer	3
- Provides an explanation of how input and output transducers are different OR - Provides an example of an input transducer and an output transducer	2
- Names devices that are transducers OR - Identifies what an input or output does	1

Question 32 (d) (ii)

Outcomes assessed: H7, H12

MARKING GUIDELINES

Criteria	Marks
Correctly evaluates R_1	2
- Recognises potential divider circuit but does not solve for R_1 OR - Carries out an equivalent calculation	1

Question 32 (d) (iii)

Outcomes assessed: H7, H12, H14

MARKING GUIDELINES

Criteria	Marks
- Calculates correct answer OR - Calculates correct R but makes mistake finding T from graph	2
- Uses correct method but makes errors in algebra so gets wrong R and $\therefore$ wrong T OR - Doesn't recognise that the resistor and thermistor are in parallel	1